

Developing simulation and statistical evaluation tools for a nonlinear mixed-effects model

Extending and validating the joint model in Leaspy

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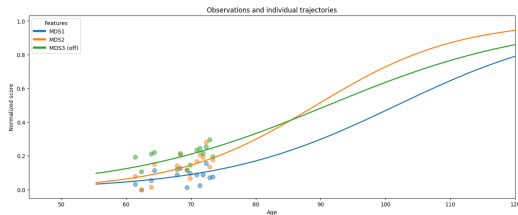
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Longitudinal data: repeated observations over time

- The **same patient** is observed at several ages.
- Visits can be sparse, irregular, unbalanced, and partially missing.
- Repeated measurements from one patient are correlated.

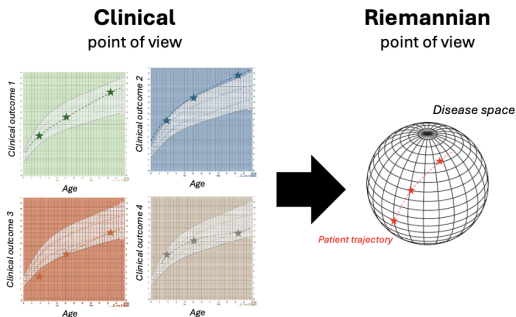


Dots: observations at successive visits. Curves: personalised trajectories.

Nonlinear mixed-effects models combine a population trajectory with patient-specific deviations.

Reconstruct disease dynamics from short, heterogeneous follow-ups.

From separate scores to a disease-space trajectory



- **Clinical view:** one curve per outcome against age.
- **Riemannian view:** their joint evolution is one path γ_i in disease space.
- The metric encodes the allowed trajectory shape.

A suitable metric on $(0, 1)^J$ produces bounded logistic trajectories.

Riemannian disease course mapping: Schiratti et al. [1].

Notation used throughout the talk

$i = 1, \dots, N$ patient

$r = 1, \dots, n_i$ visit

$j = 1, \dots, J$ longitudinal feature

$k = 1, \dots, K$ latent source

- $t_{i,r,j}$: chronological age
- $y_{i,r,j}$: observed score
- patient-specific quantities carry an index i

$$y_{i,r,j} = \gamma_{i,j}(t_{i,r,j}) + \epsilon_{i,r,j}$$

- $\gamma_{i,j}$ is the latent individual trajectory.
- $\epsilon_{i,r,j}$ represents observation noise.
- Bounded clinical scores are normalised to $(0, 1)$ for logistic modelling.

Nonlinear mixed effects: population and individual

Fixed effects: shared by the cohort

- reference time t_0
- trajectory positions $\mathbf{g}$
- trajectory velocities $\mathbf{v}_0$
- mixing matrix $\mathbf{A}$
- survival parameters in the joint model

population-average trajectory γ_0

Random effects: one set per patient

- time shift τ_i
- log-acceleration ξ_i
- latent sources $\mathbf{s}_i$
- feature shifts $\mathbf{w}_i = \mathbf{A}\mathbf{s}_i$
individual trajectory γ_i

Fixed effects describe a typical disease course; random effects explain heterogeneity.

Disease age aligns heterogeneous patients

$$\psi_i(t) = e^{\xi_i}(t - \tau_i) + t_0$$

τ_i

shifts the trajectory
earlier or later

e^{ξ_i}

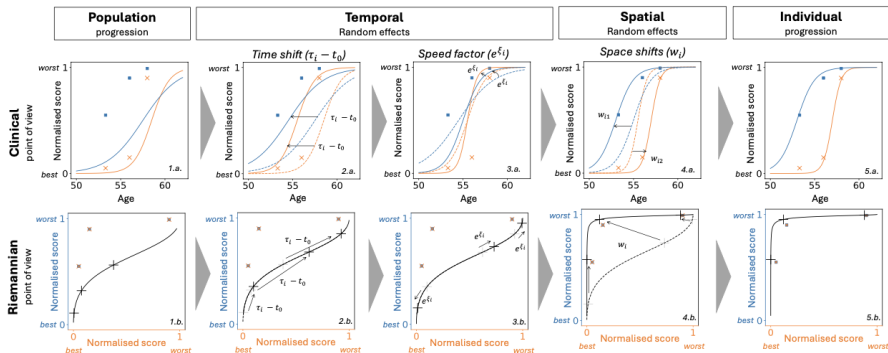
accelerates or slows
disease progression

t_0

anchors the population
disease timeline

- At chronological age $t = \tau_i$, patient i is at disease age t_0 .
- Patients of the same chronological age can therefore occupy different disease stages.

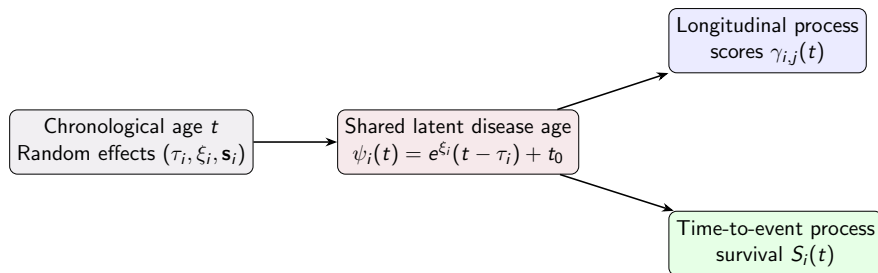
How random effects personalise the trajectory



Time shift, acceleration, and space shifts transform one population curve into an individual course.

Figure adapted from the disease course mapping presentation in Koval et al. [2].

The joint model: one disease age, two processes



- Weibull survival is applied to disease age: $S_i(t) = S_0(\psi_i(t))$.
- Latent sources can also shift the event-specific time scale.
- Supports one event or L competing events.
- Models informative dropout instead of treating it as independent missingness.

Joint latent-disease-age models: Ortholand, Durrleman, and Montcel [3] and Ortholand et al. [4].

Why simulation is indispensable here

Known truth

Real cohorts never reveal the true disease age or individual random effects.

Controlled tests

Measure recovery and probe sparse visits, censoring, and informative dropout.

Shareable cohorts

Support reproducibility and benchmarking without distributing patient records.

Internship gap. The joint-model paper used a separate research script; Leaspy's standard `simulate` interface supported only the logistic model.

Goal: integrate joint simulation in Leaspy, then test the complete simulate–fit–personalise loop.

Simulation-study principles: Morris, White, and Crowther [5].

Two input layers must remain distinct

Learned by fit

- longitudinal shape: $t_0, \mathbf{g}, \mathbf{v}_0$
- heterogeneity: σ_τ, σ_ξ
- observation noise
- survival: ν, ρ
- source mixing: $\mathbf{A}, \zeta$

Specified by the user

- cohort size N
- first-visit offset: $\bar{\delta}_f, \sigma_{\delta_f}$
- follow-up: $\bar{T}_f, \sigma_{T_f}$
- inter-visit gap: $\bar{\delta}_v, \sigma_{\delta_v}$
- minimum visit spacing
- or an explicit DataFrame visit schedule

The model determines what disease progression looks like; the user determines what the study looks like.

Simulation algorithm (1/2): create each follow-up

1. Random effects

$$\xi_i \sim \mathcal{N}(0, \sigma_\xi^2)$$

$$\tau_i \sim \mathcal{N}(t_0, \sigma_\tau^2)$$

and latent sources when
present

2. First visit

$$t_{i,0} = \tau_i + \delta_{f_i}$$

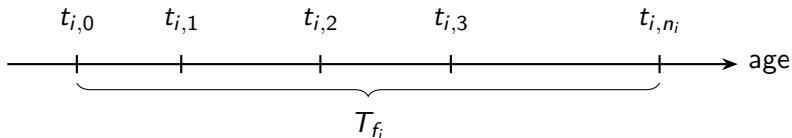
$$\delta_{f_i} \sim \mathcal{N}(\bar{\delta}_f, \sigma_{\delta_f}^2)$$

3. Visit schedule

draw inter-visit gaps

$$\delta_{v_{i,r}}$$

until the follow-up
duration T_{f_i}



Steps 2–3 are governed by the user-defined study design.

Simulation algorithm (2/2): outcomes and events

4. Outcomes

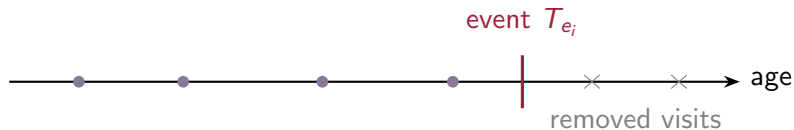
compute $\psi_i(t_{i,r})$ and γ_i
then draw bounded observations with Beta noise

5. Event time

inverse-transform sampling from survival
 $T_{e_i} \sim e^{-\xi_i} \mathcal{W}(\nu, \rho) + \tau_i$

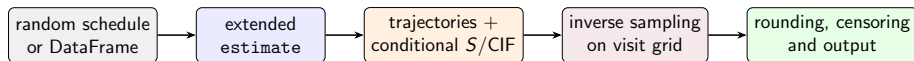
6. Censoring

remove visits after T_{e_i}
right-censor events beyond the final visit



Output: longitudinal observations, an event or censoring time, and its indicator.

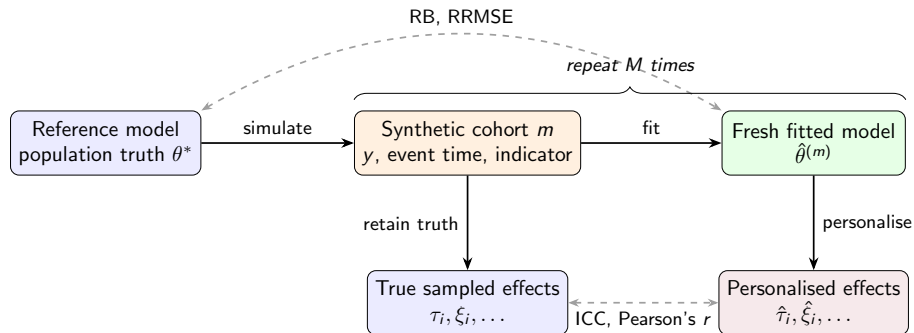
Adaptation inside Leaspy



- Missing random-mode study parameters can be estimated from an input dataset.
- Single-event survival and cause-specific CIFs use one shared interface.
- The competing-event type is sampled from incremental cause-specific CIFs.
- Events lie on the planned visit grid and cannot precede the first visit.
- Beta variance is clamped near score boundaries when necessary.
- A final pass removes visits moved beyond the event by rounding.

The implementation reuses model predictions, supports competing risks, and preserves API consistency.

How simulation quality is assessed



Self-consistency asks whether the model can recover what it generated.

What is simulated exactly?

Reference: multivariate joint model fitted on the PRO-ACT ALS dataset
 $J = 4$ longitudinal features, $K = 2$ latent sources, $L = 1$ event (death)

Bulbar function

Speech, salivation, and swallowing items.

Gross-motor function

Turning in bed; walking; climbing stairs.

Fine-motor function

Handwriting; cutting food/handling utensils; dressing and hygiene.

Total function

Overall ALSFRS-R: all items, including the preceding domains and respiratory items.

Respiratory items: dyspnea, orthopnea, and respiratory insufficiency.

Experimental setting: deliberately information-rich

Replications

$N = 1000$ patients
 $M = 200$ cohorts

Visit process

first offset -2 ± 1 y
follow-up 6 ± 2 y
interval 0.083 ± 0.042
y
minimum 0.05 y

Estimation

MCMC-SAEM
70 000 fit iterations
5 000 personalisation
iterations

Interpretation boundary: short visit intervals and long follow-up create a favourable dense-observation setting.

MCMC-SAEM for nonlinear mixed-effects models: Kuhn and Lavielle [6].

Four recovery metrics, two inference levels

Fixed effects / population parameters

Define the relative error in repetition m as

$$e_m = \frac{\hat{\theta}^{(m)} - \theta^*}{|\theta^*|} \times 100.$$

$$\text{RB} = \frac{1}{M} \sum_{m=1}^M e_m$$

signed systematic error; best value: 0%.

$$\text{RRMSE} = \sqrt{\frac{1}{M} \sum_{m=1}^M e_m^2}$$

total error from bias and dispersion; best value: 0%.

Individual random effects

ICC(3,1)

Consistency between true and personalised values while accounting for disagreement beyond rank ordering.

Pearson's r

Strength of linear association; invariant to location and scale changes.

Best value for both: 1.

Fixed effects / population parameters: recovery

Component	Parameters	Interpretation	RB (%)	RRMSE (%)
Temporal	$t_0, \sigma_\tau, \sigma_\xi$	reference and variability	0.03 to 0.44	0.83 to 2.32
Survival	ρ, ν	Weibull shape and scale	0.10 to 2.33	5.01 to 5.43
Noise	$\sigma^{(0:3)}$	observation noise	-4.60 to -1.26	1.47 to 4.73
Position	$g^{(0:3)}$	logistic position	-8.80 to -5.92	8.09 to 12.61
Velocity	$v_0^{(0:3)}$	logistic velocity	1.12 to 6.46	4.52 to 10.53

Best recovered: temporal population parameters, supported by many repeated observations per patient.

Hardest: trajectory positions g and bulbar velocity $v_0^{(0)}$, which can compensate for each other.

Individual random effects: recovery (1/2)

Effect	ICC(3,1)	Pearson's r
ξ_i	0.988 ± 0.004	0.989 ± 0.004
τ_i	0.995 ± 0.002	0.995 ± 0.002
$w_{i,0}$	0.987 ± 0.006	0.992 ± 0.003
$w_{i,1}$	0.989 ± 0.003	0.990 ± 0.003
$w_{i,2}$	0.990 ± 0.003	0.991 ± 0.003
$w_{i,3}$	0.987 ± 0.003	0.988 ± 0.003
$(\zeta^\top \mathbf{s}_i)_0$	0.965 ± 0.027	0.978 ± 0.013

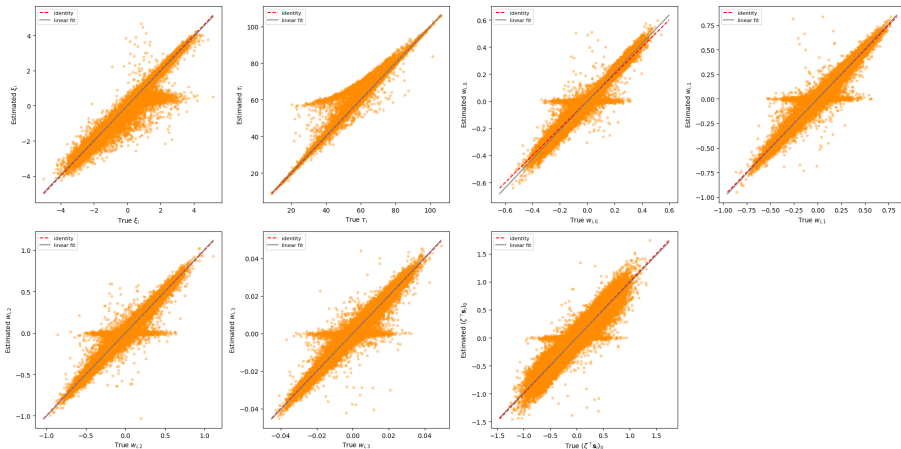
Mean $\pm$ standard deviation across repetitions.

Strong agreement for every evaluated individual effect: all ICCs exceed 0.96.

The **survival shift** is least precise: it is informed by at most one possibly censored event per patient.

Source coordinates $\mathbf{s}_i$ are rotation-dependent, so invariant feature shifts $\mathbf{w}_i = \mathbf{A}\mathbf{s}_i$ and survival shifts are compared instead.

Individual random effects: recovery (2/2)



Individual random effects: true sampled values (x-axis) vs personalised estimates (y-axis). Dashed red: identity; grey: linear fit. Points are pooled over patients and repetitions.

Take-home messages

- ① **Model:** Leaspy aligns nonlinear multivariate trajectories through an interpretable latent disease age.
- ② **Contribution:** joint simulation is integrated into the standard API, including survival, censoring, and competing events.
- ③ **Validation:** the simulate–fit–personalise pipeline is self-consistent in the favourable dense setting studied.
- ④ **Statistical lesson:** temporal and individual effects are recovered very well; trajectory positions and survival-related quantities are less precisely identified.

Simulation is both a software feature and the instrument used to validate that feature.

Patient privacy

Synthetic cohorts do not correspond to real patients and can facilitate sharing.

They still require clear labelling, provenance, governance, and disclosure-risk assessment.

Computational cost

200 fits $\times$ 70 000 iterations plus personalisation is not negligible.

Small preliminary runs and saved intermediates avoid unnecessary recomputation.

Software responsibility

Errors can propagate into scientific conclusions.

Unit tests, functional tests, documentation, and peer review are part of methodological rigour.

Synthetic does not mean risk-free, and reproducible does not mean cost-free.

References I

- [1] Jean-Baptiste Schiratti et al. “A Bayesian mixed-effects model to learn trajectories of changes from repeated manifold-valued observations”. In: *Journal of Machine Learning Research* 18.133 (2017), pp. 1–33.
- [2] Igor Koval et al. “AD Course Map charts Alzheimer’s disease progression”. In: *Scientific Reports* 11.1 (2021), p. 8020.
- [3] Juliette Ortholand, Stanley Durrleman, and Sophie Tezenas du Montcel. “A joint spatiotemporal model for multiple longitudinal markers and competing events”. In: *arXiv preprint arXiv:2501.08960* (2025).
- [4] Juliette Ortholand et al. “Joint model with latent disease age: Overcoming the need for reference time”. In: *Statistical Methods in Medical Research* 35.2 (2026), pp. 225–239.
- [5] Tim P Morris, Ian R White, and Michael J Crowther. “Using simulation studies to evaluate statistical methods”. In: *Statistics in medicine* 38.11 (2019), pp. 2074–2102.
- [6] Estelle Kuhn and Marc Lavielle. “Maximum likelihood estimation in nonlinear mixed effects models”. In: *Computational statistics & data analysis* 49.4 (2005), pp. 1020–1038.

Appendix: full longitudinal model

$$\left\{ \begin{array}{l} \psi_i(t) = e^{\xi_i}(t - \tau_i) + t_0, \\ \mathbf{w}_i = \mathbf{A}\mathbf{s}_i, \\ \gamma_{i,j}(t) = \left[1 + g_j \exp\left(-\frac{(1 + g_j)^2}{g_j} (v_{0,j}(\psi_i(t) - t_0) + w_{i,j})\right) \right]^{-1}, \\ y_{i,r,j} = \gamma_{i,j}(t_{i,r,j}) + \epsilon_{i,r,j}, \quad \epsilon_{i,r,j} \sim \mathcal{N}(0, \sigma_j^2). \end{array} \right.$$

- $p_j = 1/(1 + g_j)$ is the population trajectory value at t_0 .
- $v_{0,j}$ controls the velocity at the reference time.
- $w_{i,j}$ changes the relative ordering of feature trajectories.

Appendix: survival and latent-source coupling

Single-event survival on disease age

$$S_i(t) = \mathbf{1}_{\psi_i(t) > t_0} \exp \left[- \left(\frac{\psi_i(t) - t_0}{\nu} \right)^\rho \right] + \mathbf{1}_{\psi_i(t) \leq t_0}.$$

Event-specific scale with sources

For event l ,

$$\nu_{i,l} = \nu_l \exp \left[-\xi_i - \frac{1}{\rho_l} (\boldsymbol{\zeta}^\top \mathbf{s}_i)_l \right].$$

- Each competing event has its own Weibull pair (ν_l, ρ_l) .
- $\text{CIF}_{i,l}(t)$ gives the probability that cause l has occurred by t in the presence of competing causes.

Appendix: discrete event sampling in Leaspy

$$F_i^{\text{cond}}(t_{i,r}) = \begin{cases} 1 - S_i(t_{i,r})/S_i(t_{i,0}), & \text{single event,} \\ \sum_{l=1}^L \text{CIF}_{i,l}(t_{i,r})/S_i(t_{i,0}), & \text{competing events.} \end{cases}$$

- 1 Draw $U \sim \mathcal{U}[0, 1]$.
- 2 Select the first planned visit where $F_i^{\text{cond}}(t_{i,r}) \geq U$.
- 3 If no visit crosses U , right-censor at the last visit.
- 4 For competing risks, sample the cause from incremental cause-specific CIFs.

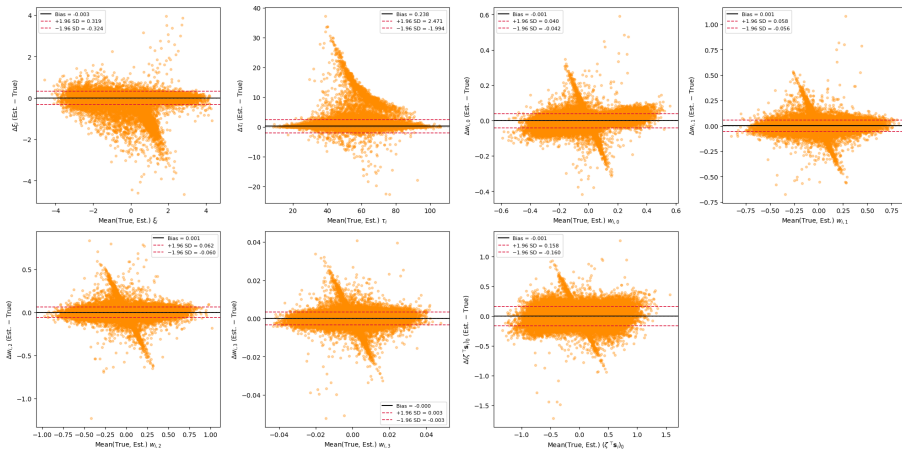
This ensures internal API reuse, but discretises event times onto the visit grid.

Appendix: exact population recovery values

Parameter	RB (%)	RRMSE (%)	Parameter	RB (%)	RRMSE (%)
$g^{(0)}$	-8.27	11.41	$v_0^{(0)}$	6.46	10.53
$g^{(1)}$	-8.80	12.61	$v_0^{(1)}$	2.23	5.57
$g^{(2)}$	-7.60	11.16	$v_0^{(2)}$	1.12	4.52
$g^{(3)}$	-5.92	8.09	$v_0^{(3)}$	2.05	5.25
ρ	0.10	5.01	ν	2.33	5.43
$\sigma^{(0)}$	-4.60	4.73	$\sigma^{(1)}$	-4.16	4.25
$\sigma^{(2)}$	-3.08	3.17	$\sigma^{(3)}$	-1.26	1.47
t_0	0.44	0.83	σ_τ	0.08	2.32
σ_ξ	0.03	2.32			

The appendix reports the exact values; the main slide groups them by model component for readability.

Appendix: Bland–Altman diagnostic



Black: mean difference; red: limits of agreement. Biases are close to zero for all effects.

Simulation unit tests

- non-empty random-mode output
- expected patient count
- strictly increasing visit times
- DataFrame schedules are respected
- all model features are returned

estimate functional tests

- univariate numerical correctness
- raw/DataFrame consistency
- MultiIndex preservation
- multivariate output shape:
 $n_{\text{timepoints}} \times (J + L)$

Appendix: limitations and next experiments

What the reported experiment establishes

Self-consistency of the integrated pipeline for $N = 1000$ under a favourable, dense visit design.

What it does not establish

Universal recovery under sparse visits, heavier censoring, smaller cohorts, or model misspecification.

Perspectives stated in the report

Vary sample size, censoring intensity, visit sparsity, and misspecification; explore more flexible visit schedules and noise models; align continuous-time event generation more closely with the grid-based implementation.